

Quantum Physical Unclonable Functions: Foundations and Applications

Tutorial Abstract — ECOC 2026, Málaga

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Abstract. Quantum Physical Unclonable Functions extend the no-cloning principle from states to transformations, enabling hardware-rooted security for quantum networks. This tutorial introduces foundations, security notions, impossibility and positive results, and applications to authentication, identification, hybrid PUFs, and practical quantum communication infrastructures, with implementation considerations and outlook.

Keywords: Quantum physical unclonable functions; no-cloning; unclonability of unitaries; quantum authentication; quantum networks; hardware security; hybrid PUFs.

Motivation and scope

As quantum communication systems move from demonstrations to networked infrastructure, security must be formulated not only for abstract protocols but also for the physical devices that implement them. Classical cryptography and post-quantum assumptions remain essential, yet they usually treat devices as endpoints that store keys or execute algorithms. Quantum networks expose a richer attack surface: an adversary may interact coherently with a device, collect quantum challenge-response data, exploit noise or drift, or try to emulate the device rather than recover a digital secret. This tutorial introduces quantum physical unclonable functions (QPUFs) as a hardware-rooted security paradigm for this setting, building on the formal QPUF framework and the broader programme of quantum unclonability and cryptanalysis [1, 2].

From no-cloning to unclonability of transformations

The tutorial opens with a quantum-native foundation: the route from the no-cloning theorem [3] to extended notions of unclonability. For device security, the protected object is often not a single state but an unknown optical scattering process, unitary evolution, noisy channel, or device-specific map from quantum challenges to quantum responses. The operational question behind QPUFs is therefore: after a limited number of interactions with a physical device, can an adversary build another device, circuit, or strategy that behaves like it on future challenges?

This shift changes the security problem from copying states to learning, emulating, cloning, or forging transformations. The tutorial will distinguish copying an unknown state, reproducing stored challenge-response pairs, predicting selected future responses, and constructing a reusable emulator for an unknown unitary. These distinctions are essential for understanding why QPUFs are not merely “classical PUFs with qubits”, but an attempt to formalise physical unclonability as a cryptographic resource for quantum processes [1, 4].

Formal foundations and security hierarchy

In its cleanest mathematical form, a QPUF is modelled as an unknown unitary transformation acting on quantum states: challenges are quantum states and responses are transformed output states. The cryptographic question is whether an adversary, after training access to the device, can answer a fresh verification challenge with sufficient fidelity to be accepted. This gives rise to a hierarchy of notions, including selective unforgeability, existential unforgeability, and stronger emulation-style security [1, 4].

A central message is that quantum mechanics provides powerful resources but does not make every security goal possible. General quantum emulation attacks rule out overly strong forms of QPUF security, while unitary QPUFs can satisfy the weaker and application-relevant notion of selective unforgeability. The tutorial will also explain how QPUF security connects to quantum pseudorandom states and unitaries [5], and how process-learning results clarify which device families are vulnerable to statistical-query style attacks [6, 7].

Applications to authentication and quantum networks

Authentication is a natural application because many quantum network primitives, including quantum key distribution, assume authenticated channels. QPUFs offer a route in which the trust anchor is a physical quantum transformation rather than a long-term stored secret. The tutorial will present client-server identification protocols in which a verifier prepares quantum challenges and checks whether the prover can return the corresponding device-specific responses [8]. Different architectures allow either the client or the server to be the high-resource quantum party.

The tutorial will also discuss hybrid and entanglement-assisted routes that are closer to near-term quantum optical systems. Hybrid locked PUFs encode classical PUF outputs into non-orthogonal quantum states, limiting the information available to an adversary and mitigating machine-learning attacks against classical strong PUFs [9]. More recent entanglement-based authentication protocols combine weak classical PUFs with local indistinguishability and introduce the Hybrid Entangled PUF (HEPUF) module for advanced quantum networks [10]. These constructions show that quantum hardware security need not rely only on a perfect unknown unitary; it can also exploit non-orthogonality, measurement disturbance, limited accessibility, and entanglement.

References

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